南 京 林 业 大 学

Nanjing Forestry University

硕士研究生开题报告

Master's Thesis Opening Report

Latency-Aware Neuro-Symbolic Path-Tracking Control for Autonomous Vehicles

作者 / Author	Rizwan Ali
学号 / Student No.	_____
专业 / Major	_____
研究方向 / Research Direction	Autonomous Systems
指导老师 / Supervisor	Jie Tian

25 October 2026

目录 / Contents

1 课题名称、来源及选题意义 / Thesis Title, Source and Significance **3**

2 国内外研究现状 / Review of Related Research **4**

3 研究目标及内容 / Research Objectives and Contents **6**

4 研究方法与技术路线 / Research Methods and Technical Route **11**

5 拟议创新点 / Proposed Contributions **16**

6 可能问题及解决措施 / Potential Problems and Responses **17**

7 应用价值 / Application Value **17**

8 总体安排及进度 / Research Arrangement and Schedule **18**

9 经费预算 / Proposed Budget **18**

参考文献 / References **20**

Abbreviations

LANS: Latency-Aware Neuro-Symbolic System
LQR: Linear Quadratic Regulator
PPO: Proximal Policy Optimization
MPC: Model Predictive Control
RMSE: Root Mean Square Error
RK4: Fourth-Order Runge–Kutta

1 课题名称、来源及选题意义 / Thesis Title, Source and Significance

1.1 Thesis title

Latency-Aware Neuro-Symbolic Path-Tracking Control for Autonomous Vehicles

1.2 Research source and scope

This master's research at Nanjing Forestry University investigates the Latency-Aware Neuro-Symbolic System (LANS), a path-tracking framework that combines an analytical steering controller with selectively admitted learned corrections. The work is supervised by Jie Tian. A speed-scheduled linear quadratic regulator (LQR) provides the baseline command. A learned policy proposes a bounded residual, while explicit runtime rules and a nominal predictive test determine whether that residual receives control authority.

In this report, "neuro-symbolic" denotes the combination of a neural proposal mechanism and inspectable supervisory rules. It does not imply learned logical reasoning or a formal safety certificate. The present research platform uses nonlinear vehicle simulation, state estimation, constrained steering actuation and timestamped action delivery. Physical-vehicle deployment is outside the confirmed scope.

The proposal builds on the retained supervised student and the audited execution interface. Proximal policy optimization (PPO) remains a candidate learning method and a conditional follow-up experiment; it is not presented as the current best-performing policy. The repository baseline reviewed for this report is commit `17dbd78` [17–19].

1.3 Background and significance

Path tracking depends on both steering quality and the age of the information used to calculate a command. A numerically valid neural output can become unsuitable while waiting for computation or delivery. Conversely, a fresh output may still degrade a strong analytical controller. The research therefore treats freshness and usefulness as separate admission conditions.

LQR offers a transparent local reference with modest computation, but its nominal model cannot fully represent nonlinear tire behavior, parameter mismatch and actuator dynamics. Learned residuals offer a way to adapt corrections without replacing the analytical baseline. Their benefit must nevertheless be demonstrated against the same strong baseline and against nonlearned corrections passed through the same supervisor. A hybrid improvement alone does not establish that learning caused the gain.

The central question is: **can a learned residual earn useful control authority under delayed execution while preserving the analytical command whenever the proposal is stale, invalid or predictively unhelpful?** The scientific objective is to separate the contributions of the proposal, predictor, objective and execution rules. The engineering objective is to make those rules reproducible and compatible with a scheduled local controller.

1.4 Preliminary foundation and remaining research

Recorded evaluations provide a modest research foundation. In the revision9 fresh-noise comparison, the frozen supervised student with the retained gate achieved mean penalized episode RMSE of 0.082563 m, compared with 0.084307 m for LQR, a reported reduction of 2.068%. Both completed 108 of 108 cases. A fixed −0.04 rad proposal through the same gate achieved 0.083617 m. These are known-path simulation results, not evidence of unseen-road generalization [17].

PPO fine-tuning did not improve this retained student. All three development selections retained step zero; the 100,000-step fine-tuned endpoints had 0.323% higher mean error than the student on the recorded final comparison. These outcomes motivate a diagnosis of gate prediction and objective choice before further training. They do not support a claim of PPO dominance or a general 5–10% gain.

The audit corrected interface and evaluation defects and identified limits in native transfer [18]. Revision12 freezes a 1,080-episode development ablation of the retained predictor, curvature-preview prediction and a lateral-only objective [19]. Its code and protocol are available; no completed revision12 result was present in the reviewed checkout. The remaining work starts with this bounded evaluation and proceeds to additional learning only if its declared conditions pass.

2 国内外研究现状 / Review of Related Research

2.1 Classical vehicle path tracking

Vehicle dynamics and optimal-control literature establish the modeling assumptions behind linear feedback and predictive control [1,2]. LQR synthesizes gains for a stated linear model and quadratic objective; its guarantees do not automatically extend to a nonlinear vehicle with saturation, delayed proposals and supervisory switching. Model predictive control (MPC) offers explicit horizon-based handling of constraints, but its performance and computational demands depend on model fidelity, tuning, information and solver settings [12].

For this thesis, LQR and any MPC reference must share the relevant physical plant, sensing, delay and actuator conditions. A deliberately weak baseline would make an apparent learning benefit difficult to interpret. The retained speed schedule and development-selected LQR weights will therefore be frozen before comparative evaluation.

2.2 Learned and residual control

Deep reinforcement learning has been studied across autonomous-driving tasks, although success in a particular simulated maneuver does not establish transferable vehicle-control performance [3]. Residual reinforcement learning augments an existing controller with a learned correction [4]. That formulation is established prior work and is not itself claimed as a contribution of this thesis.

PPO, generalized advantage estimation and reproducible software implementations provide

practical training components [5–7]. A residual policy and a full-command policy have different action semantics and require separately identified checkpoints. Supervised learning from a bounded teacher can also supply a residual policy. The retained LANS student was obtained through that route; comparisons with PPO must distinguish the effect of initialization, pretraining data and additional optimization budget.

2.3 Timing, delay and information age

Real-time and networked-control research shows that scheduling, communication and delayed feedback affect closed-loop behavior [8,9]. Inference duration measures time spent computing an output. Information age measures elapsed time since the associated observation was sampled. Neither quantity should be substituted for the other, and neither alone determines the command's predicted benefit.

The current LANS surrogate uses a deterministic service-delay scheduler. It supports controlled action-age experiments but does not establish operating-system scheduling performance. A separate process-based implementation is needed to evaluate actual inference contention, local-loop jitter and deadline misses. Hardware measurements and simulation-clock measurements must remain distinct.

2.4 Runtime supervision and assurance

Predictive safety filters and safe-learning methods explicitly state the assumptions needed for their constraint and safety results [10,11]. LANS instead applies a nominal performance gate: it compares predicted costs of candidate fused commands with that of LQR and combines that comparison with age, validity, sequence and envelope checks. This does not establish invariance, recursive feasibility, collision avoidance or safety outside the assessed domain.

"Fallback to LQR" means that the selected steering command equals the analytical baseline when a proposal is rejected. It does not mean that LQR can recover from every state. Likewise, measured execution-time maxima are observations from a finite campaign, not verified worst-case execution-time bounds [15].

2.5 Research gap and comparative position

The working gap is the empirical attribution of small residual gains under delayed execution: whether improvement comes from the learned proposal, the supervisor's selection of any useful correction, or differences in timing and model assumptions. The proposed contribution is an explicit, tested integration and an auditable evaluation methodology, subject to comparison with the closest residual-control, delay-aware and predictive-filtering methods.

Method family	Established capability	Question for LANS
Residual learning [4]	Learned correction around a baseline	Does learning outperform fixed corrections under the same gate?

Method family	Established capability	Question for LANS
Networked/real-time control [8,9]	Analysis of timing and delayed information	Which measured ages and execution states determine useful authority?
Predictive filtering [10,11]	Model-based intervention with stated assumptions	How often does nominal cost ordering agree with realized benefit?
Constrained MPC [12]	Prediction and constraints in command generation	Does any advantage survive equal-information and compute-aware comparison?

Multiple seeds, frozen selection rules and honest uncertainty reporting are required [13,14]. Existing work is not assumed to omit all LANS features, and the combination of familiar components is not treated as proof of algorithmic novelty. Full-text comparison and, where feasible, matched reproduction remain part of the thesis work.

3 研究目标及内容 / Research Objectives and Contents

3.1 Research objectives

The overall objective is to evaluate a latency- and freshness-aware hybrid controller in which a strong LQR baseline retains default authority and a learned residual is admitted only when its actual fused command passes explicit supervisory conditions.

1. Freeze consistent coordinates, steering units, sensing assumptions, actuator limits and packet semantics across comparisons.
2. Quantify the retained student's benefit relative to LQR and same-gate fixed residuals.
3. Isolate predictor and objective changes before spending on additional policy training.
4. Evaluate action age separately from inference duration and measure the effects of packet holding, rejection and recovery.
5. Establish whether conditional PPO fine-tuning adds benefit beyond the supervised initialization.
6. Prepare a causal native observation interface and assess actual process timing before claiming transfer to a 100 Hz local loop.
7. Retain failed outcomes, report uncertainty and provide a reproducible thesis evidence package.

3.2 Vehicle dynamics and coordinate conventions

The evaluation plant uses a nonlinear bicycle model. Its world coordinates are X, Y, heading is ψ, prescribed longitudinal speed is $v_x > 0$, lateral velocity is v_y, yaw rate is r, and actual front-wheel steering is δ_a. Positive lateral force and yaw point to the vehicle's left. Longitudinal speed regulation is outside the steering study.

$$\dot{X} = v_x \cos\psi - v_y \sin\psi, \qquad \dot{Y} = v_x \sin\psi + v_y \cos\psi, \qquad \dot{\psi} = r. \tag{1}$$

$$m(\dot{v}_y + v_x r) = F_{yf} \cos\delta_a + F_{yr}, \qquad I_z \dot{r} = l_f F_{yf} \cos\delta_a - l_r F_{yr}. \tag{2}$$

$$\alpha_f = \delta_a - \arctan\left(\frac{v_y + l_f r}{v_x}\right), \qquad \alpha_r = -\arctan\left(\frac{v_y - l_r r}{v_x}\right). \tag{3}$$

For either axle, let $q = \tan\alpha$, cornering stiffness be C, normal load be F_z and friction coefficient be μ. The implemented Fiala-type relation is

$$F_y = \begin{cases} Cq - \dfrac{C^2}{3\mu F_z}|q|q + \dfrac{C^3}{27\mu^2 F_z^2}q^3, & |q| < 3\mu F_z/C, \\ \mu F_z \operatorname{sgn}(q), & \text{otherwise.} \end{cases} \tag{4}$$

Nominal parameters are $m = 1650$ kg, $I_z = 4224$ kg m², $l_f = 1.2$ m, $l_r = 1.4$ m and $C_f = C_r = 80000$ N/rad. Static axle loads are $F_{zf} = mgl_r/(l_f+l_r)$ and $F_{zr} = mgl_f/(l_f+l_r)$. These are simulation parameters, not verified specifications of a named production vehicle. Load transfer and a separately modeled longitudinal controller are not included.

To reproduce current evidence, a path is represented as $Y_{ref}(X)$. The implemented errors and local curvature are

$$e_y = Y - Y_{ref}(X), \quad e_\psi = \psi - \arctan Y'_{ref}(X), \quad \kappa = \frac{Y''_{ref}(X)}{[1 + (Y'_{ref}(X))^2]^{3/2}}. \tag{5}$$

Thus e_y is a world-coordinate vertical displacement, not an exact closest-point perpendicular distance. The local approximations $\dot{e}_y \approx v_y + v_x e_\psi$ and $\dot{e}_\psi \approx r - v_x\kappa$ support the linear design. A future Frenet-coordinate implementation would require a separate interface revision and new evaluation; it must not be substituted into historical results without revalidation.

3.3 State estimation and the analytical baseline

The observer estimates $x = [e_y, v_y, e_\psi, r]^T$ from noisy lateral-error, heading-error and yaw-rate measurements, with measured speed and available actuator angle. It does not directly measure lateral velocity. Map localization, curvature previews and actuator state are idealized available inputs in the current surrogate. Their availability and errors must be reassessed in a native implementation.

At a fixed design speed v, the nominal linear model is

$$\dot{x} = A(v)x + B\delta_a + d_\kappa, \qquad d_\kappa = [0, 0, -v\kappa, 0]^T. \tag{6}$$

$$A(v)=\begin{bmatrix}0 & 1 & v & 0\\ 0 & -\frac{C_f+C_r}{mv} & 0 & \frac{C_r l_r-C_f l_f}{mv}-v\\ 0 & 0 & 0 & 1\\ 0 & \frac{C_r l_r-C_f l_f}{I_z v} & 0 & -\frac{C_f l_f^2+C_r l_r^2}{I_z v}\end{bmatrix},\quad B=\begin{bmatrix}0\\ C_f/m\\ 0\\ C_f l_f/I_z\end{bmatrix}. \tag{7}$$

The LQR gain satisfies $K(v)=R^{-1}B^TP(v)$, where P solves the continuous-time algebraic Riccati equation. The retained comparison uses $Q=\mathrm{diag}(4,0.1,8,0.2)$ and $R=100$, with gains calculated at 15, 20 and 25 m/s and interpolated for measured speed. These values belong to the retained controller; they are not retuned using confirmation outcomes.

$$\delta_{LQR}=\mathrm{clip}\big(\delta_{ff}-K(v)(\hat{x}-x_{ss}),-\delta_{max},\delta_{max}\big). \tag{8}$$

With $L=l_f+l_r$, the nominal steady-state terms are $v_{y,ss}=(l_r v-m l_f v^3/(LC_r))\kappa$, $x_{ss}=[0,v_{y,ss},-v_{y,ss}/v,v\kappa]^T$ and $\delta_{ff}=[L+(m/L)(l_r/C_f-l_f/C_r)v^2]\kappa$. This makes reference-curvature forcing and feedforward explicit. The same baseline is used for the hybrid and the LQR comparison.

3.4 Learned residuals and observation semantics

The retained student uses a 20-dimensional observation, a train-only standardizer, two 64-unit tanh hidden layers and five output logits. Deterministic evaluation selects one of five residuals. A full-command PPO policy is a distinct comparator; it must not be relabeled as a residual policy.

$$\begin{gathered}\eta_t=[o_t^T,c_t^T]^T\in\mathbb{R}^{20},\quad \tilde{\eta}_t=(\eta_t-\bar{\eta}_{train})/s_{train},\\ \Delta\delta_t\in\{-0.08,-0.04,0,0.04,0.08\}\ \text{rad}.\end{gathered} \tag{9}$$

In Figure 1, the observation builder provides nine tracking, speed, actuator and curvature-preview channels in o_t. The context c_t supplies ten gate, held-packet, pending-worker and previous-command channels plus one configured service-delay channel. The policy standardizes their 20-dimensional concatenation η_t. Curvature previews correspond to 0, 10 and 20 m ahead along the current map coordinate. Missing-packet indicators are explicit and numerical features remain finite.

An important transfer limitation is that the simulator exposes a pending proposal and its scheduled remaining service time. Those values are known to the scheduler but are not necessarily observable before a real inference worker completes. Native deployment therefore requires an observation audit: unavailable values must be removed or causally estimated, the feature contract versioned, and affected policies retrained and reevaluated. Replacing unavailable values silently at deployment would invalidate an equivalence claim.

The supervised teacher used privileged current-state and parameter information for offline labels. It did not provide actual future noise to the student. Training-only privileged calculations must remain outside the deployed policy and gate. Conditional PPO studies will use the same executed-command path and distinguish added RL budget from the student's pretraining budget.

3.5 Timestamped delivery and freshness

Each accepted request is associated with an observation timestamp and sequence identifier. One request may be in flight; the most recent delivered packet can be held between completions. Holding a packet does not reset its timestamp. The current surrogate evaluates on a 10 ms control grid with configured service delays; it does not equate those delays with measured CPU inference duration.

$$\ell_j = t_{finish,j} - t_{start,j}, \qquad a_j(t) = t - t_{obs,j}. \qquad (10)$$

Here ℓ_j is wall inference duration only when both timestamps come from the same monotonic wall clock. The retained gate's a_j is age on the simulation control clock. Physical sensor age can exceed that scheduler age when input measurements are already delayed. A native packet interface must record acquisition, request, start, completion and use times consistently and report any additional command-delivery delay.

The retained scheduler rejects missing packets, nonfinite data, negative age, age of at least 0.05 s, and sequence regressions. Reuse of the same delivered sequence is allowed. Strict duplicate-arrival handling, policy-version checks and acquisition-level sensor timestamps are sensible native extensions, but are not described as already validated features of the current scheduler.

3.6 Predictive symbolic supervisor and fusion

The supervisor receives the state estimate, current analytical command, measured speed, local curvature, actuator angle and delivered residual. It first applies validity, age, sequence and tracking-envelope checks. The retained learned-control envelope is $|\widehat{e}_y| \le 1.5$ m and $|\widehat{e}_\psi| \le 0.25$ rad. This is an admission envelope, not a proven invariant set or an LQR recovery domain.

$$\delta_{cand} = \mathrm{clip}\big(\delta_{LQR} + \mathrm{clip}(\Delta\delta, -0.08, 0.08), -0.5, 0.5\big)\,. \qquad (11)$$

$$\begin{aligned} f(a) &= \mathrm{clip}\left(\frac{0.05 - a}{0.04}, 0, 1\right), \\ w^{(b)} &= \min\{b f(a), w_{prev} + 0.05\}, \quad b \in \{0.10, 0.25, 0.50\}. \end{aligned} \qquad (12)$$

For each candidate weight, the gate evaluates the actual fused command $u^{(b)} = (1 - w^{(b)})\delta_{LQR} + w^{(b)}\delta_{cand}$. The candidate with minimum finite predicted cost is compared with the pure-LQR branch. Weight selection therefore depends on predicted benefit, freshness and previous authority; it is not a function of inference duration alone.

The retained predictor uses the nominal linear bicycle and rate/lag actuator over 20 steps of 0.01 s. A tested command is held for the first five steps, followed by the same local-curvature LQR continuation in both branches. With $h = 0.01$ s, its stated dimensionless scaling gives

$$J(u) = h\sum_{i=1}^{20}\left[\left(\frac{e_{y,i}}{0.5}\right)^2 + 0.2\left(\frac{e_{\psi,i}}{0.1}\right)^2 + 0.01\left(\frac{r_i - v\kappa}{0.2}\right)^2 + 0.001\left(\frac{u_i}{0.5}\right)^2\right] + J_f. \qquad (13)$$

$$J_f = 0.1\left[\left(\frac{e_{y,20}}{0.5}\right)^2 + 0.2\left(\frac{e_{\psi,20}}{0.1}\right)^2\right], \qquad J_{LQR} - J_{best} \geq \max(10^{-7}, 0.01J_{LQR}). \qquad (14)$$

A positive weight and three consecutive qualifying control ticks are required before authority is applied. These are qualifying ticks, not necessarily three distinct packets. Buffered reuse can advance persistence if all conditions continue to pass. Rejection resets both weight and persistence. Once enabled, upward weight growth is limited to 0.05 per tick; rejection can set the weight to zero immediately.

$$\delta_{cmd} = \begin{cases} (1-w)\delta_{LQR} + w\delta_{cand}, & \text{admitted,} \\ \delta_{LQR}, & \text{missing, invalid, stale or rejected.} \end{cases} \qquad (15)$$

Fallback uses an explicit branch so that invalid residual arithmetic is not evaluated. Physical steering remains constrained by the shared actuator. The software gate computes the selected command; the separate Fusion Layer in Figure 1 is a conceptual presentation of the final algebra, not a second addition of the residual.

3.7 Actuator dynamics and execution limits

The nonlinear plant integrates at 1 ms using fourth-order Runge–Kutta while control updates every 10 ms. All methods use the same actuator:

$$\dot{\delta}_a = \text{clip}\left(\frac{\text{clip}(\delta_{cmd}, -0.5, 0.5) - \delta_a}{0.05}, -0.35, 0.35\right), \qquad |\delta_a| \leq 0.5. \qquad (16)$$

Angles are in radians, angular rates in rad/s and time in seconds. A future command-transmission queue would replace the actuator's command input with the delivered command for every comparator. Such a queue is an extension, not an assumed element of the current revision8/9 equations.

Real-process inference will use a nonblocking local loop and bounded work. LQR must be computed before optional prediction. A timeout fallback cannot be guaranteed merely by checking elapsed time after a blocking predictor returns: bounded iterations or an interruptible architecture and measured local deadlines are required. A 100 Hz simulation schedule is not a hard-real-time guarantee.

3.8 Research outputs

The thesis will deliver a versioned controller specification, verified baseline and execution interfaces, comparative results, a predictor/objective attribution study, conditional policy-learning results and a reproducibility package. A documented negative result is an acceptable outcome. Further training, a revised gate or native deployment will not be described as completed before its evidence exists.

4 研究方法与技术路线 / Research Methods and Technical Route

4.1 Verification and evidence management

The first stage consolidates audited dynamics, steering signs, observation order, reference conventions, packet timestamps and actuator behavior. Regression checks cover candidate construction, single fusion, fallback equality, persistence, delayed delivery and consistent training/evaluation execution. Integration-step sensitivity and small-slip tire behavior will also be checked. Historical outputs affected by identified defects remain excluded from corrected aggregate claims [18].

For each experiment, code, policy, normalizer, parameters and noise identifiers will be frozen before evaluation. Replaying identical observations, arrivals and proposals through relevant wrappers will test execution agreement. Checks that exercise code contracts will be distinguished from closed-loop performance measurements.

4.2 Technical route and framework

The sequence is: retained-baseline verification; literature and interface review; predictor/objective ablation; independent teacher-headroom confirmation when eligible; bounded PPO pilots when justified; frozen fresh-noise evaluation; and separate native-interface and timing assessment.

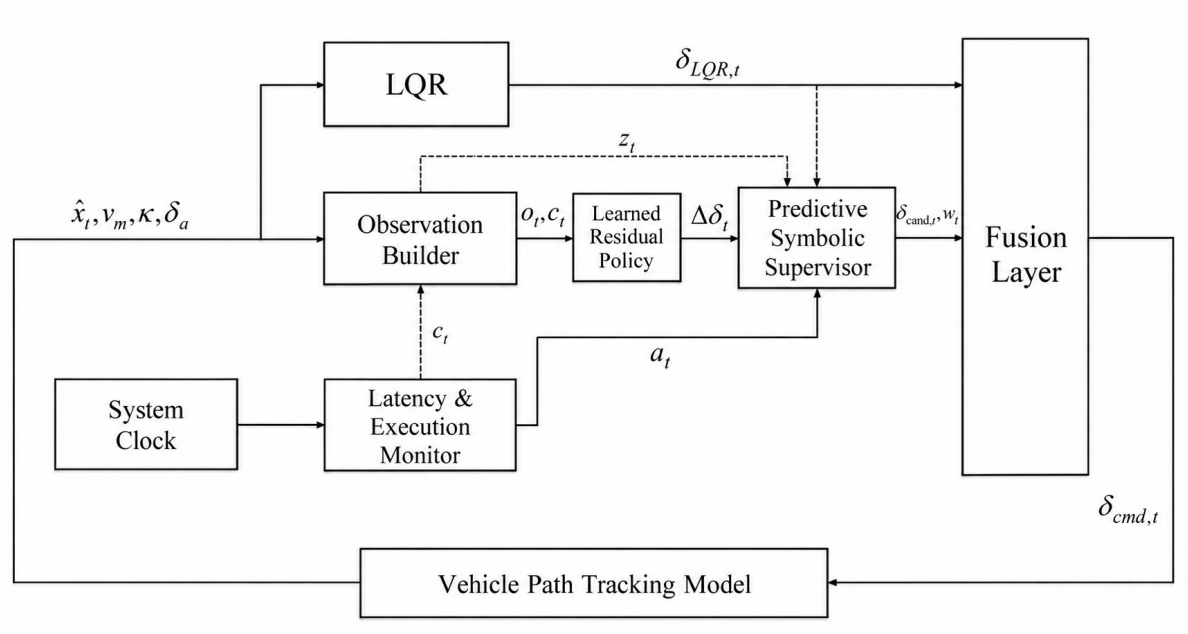


Figure 1. Final LANS framework supplied by the author: an LQR baseline, an observation builder, a learned residual policy, a predictive symbolic supervisor, a latency and execution monitor, a fusion layer and the vehicle path-tracking model.

Figure 1 uses conceptual blocks. The observation builder passes current state and available reference/actuator information to the supervisor through z_t, while the LQR branch also supplies its baseline command. The policy receives the tracking/preview features o_t and execution context c_t; their concatenation forms the 20-feature input specified in Section 3.4. The current policy is the retained supervised student, with PPO reserved for the conditional learning study.

Figure symbol	Definition in this report
$\hat{x}_t, v_m, \kappa, \delta_a$	Estimated tracking state, measured speed, local path curvature and available actuator steering.
o_t, c_t	Nine physical/preview channels and eleven execution/service-context channels; concatenated before policy standardization.
z_t	Current supervisory information assembled from available estimates, reference and actuator signals; it is not a second learned output.
$\Delta\delta_t$	Delivered bounded residual proposal associated with its original observation timestamp.
a_t	Age of the delivered proposal, $t - t_{obs}$, using a consistent clock.
$\delta_{LQR,t}$	Complete analytical baseline command at the current control tick.

Figure symbol	Definition in this report
$\delta_{cand,t}, w_t$	Clipped full candidate command and selected blend weight.
$\delta_{cmd,t}$	Final command selected by Equation (15) and applied through the shared actuator.

The simple schematic omits packet-buffer internals and timestamp/history exchange with the monitor. Those exchanges remain required by Sections 3.4–3.6; the monitor cannot determine action age from the system clock alone. Reference and sensing preprocessing are abstracted into the feedback input bundle. The vehicle path-tracking model includes the shared steering actuator described in Section 3.7. The separate Fusion Layer illustrates the gate's final algebra and must not add the residual a second time. Native causal-interface revisions remain proposed work.

4.3 Predictor and objective ablation

Revision12 compares three frozen gate variants: the retained constant-curvature predictor and combined cost; a preview-only variant that changes predicted curvature using the existing preview channels; and a lateral-objective-only variant that retains the original dynamics but removes the heading, yaw and steering-effort terms from its scalar objective [19]. All share the same plant, observations, actuator, packet lifecycle, rejection logic, persistence and weight ramp.

Each gate receives the frozen student and the two predeclared constant proposals −0.04 and +0.04 rad. Pure LQR is the common reference. Three student/noise pairs, three speeds, two paths, six condition families and ten methods produce 1,080 development episodes. No policy retraining is needed for this first attribution stage.

Eligibility requires both gate progress and additional learned benefit. Gate progress requires at least 1% overall improvement over the retained-gate student, improvement in all three paired seed means and at least two declared stress regions, plus preservation checks. Learned benefit requires at least 1% overall improvement over both constants through the same gate, without cellwise survival loss. The full protocol also bounds active-region regression, preserves nominal tracking and requires exact severe-delay fallback.

Only an eligible gate proceeds to the independent 288-episode teacher-headroom check. Only a passing teacher check permits two 100,000-step PPO pilots. A selected gate and retained policies must be frozen before the conditional 432-episode fresh-noise evaluation. Failed gate or teacher screens stop that path and leave reserved noise unused. A 5% regional gain is a separately assessed target, not an achieved result or a mandatory reporting threshold. Changes to the frozen decision tree require a new protocol.

4.4 Baselines and controlled comparisons

ID	Controller or intervention	Purpose and status
B1	Retained speed-scheduled LQR	Shared analytical reference.
B2	Frozen supervised student with retained gate	Current retained research controller.
B3	Fixed −0.04 and +0.04 rad under each gate	Attribute benefit beyond nonlearned proposals.
B4	Frozen student with preview-only or lateral-only gate	Isolate the revision12 gate changes.
B5	Same policy with bounded gate interventions	Diagnose freshness/performance/authority; separate protocol if not in revision12.
B6	Student-initialized PPO endpoints	Conditional learning study, compared with its own frozen initialization.
B7	Matched standalone or direct-residual PPO	Secondary comparator requiring distinct training and action semantics.
B8	Process-based versus simulated execution	Secondary timing study after a causal interface is validated.

A constrained, actuator-aware MPC may be included as a secondary reference when its sensing, delays, information and compute budget are documented. The earlier nominal MPC diagnostic is not evidence against all well-tuned MPC designs. Frozen-policy interventions and independently retrained variants will be labeled separately.

4.5 Scenarios, disturbances and feasibility

The retained development paths are sine120 and DLC220 at 15, 20 and 25 m/s. Fresh noise on these geometries is not unseen-road testing; all three speeds are represented in existing development. Additional paths or speed ranges would require separately reserved generalization tests. DLC220 is a research geometry and is not claimed to satisfy ISO 3888-1.

Condition family	Declared setting in the retained protocol
Nominal	Friction 0.9; nominal tire stiffness and sensor noise; zero service delay.
Tire mismatch	Plant stiffness multiplier 0.65; nominal predictor unchanged.
Low friction	Plant friction 0.6; nominal predictor unchanged.

Condition family	Declared setting in the retained protocol
Sensor stress	Twice nominal noise and one control-step sensor delay.
Latency20	20 ms simulated service delay for learned/nonlearned proposals.
Latency100	100 ms service delay; stale-packet rejection must reproduce LQR.

Nominal measurement-noise standard deviations are 0.03 m for lateral error, 0.17° for heading error, 0.01 rad/s for yaw rate and 0.1 m/s for speed. The small-angle observation calculations, ideal map localization and available actuator steering remain stated assumptions. Physical mismatch and sensor delay are not silently folded into inference timing.

Feasibility screening will inspect curvature, demanded lateral acceleration, tire friction and steering/rate limits before additional path-speed combinations are frozen. Confirmed infeasible cases will be labeled explicitly rather than removed after comparative results are inspected. Worker outage, reordering and acquisition-age stress are separate planned interface tests, not already covered by the three deterministic service-delay values.

4.6 Metrics, uncertainty and decision rules

The primary comparison uses the equally weighted mean of episode-level penalized RMSE, not a pooled mean over all time samples. For a scheduled horizon of N samples, n executed samples and the retained lateral failure penalty of 2.5 m,

$$\mathrm{RMSE}_{pen} = \sqrt{\frac{\sum_{k=1}^{n} e_{y,k}^2 + (N-n)(2.5)^2}{N}}, \qquad \bar{E} = \frac{1}{M}\sum_{j=1}^{M} \mathrm{RMSE}_{pen,j}. \tag{17}$$

$$G_{A\to B} = 100\left(1 - \frac{\bar{E}_B}{\bar{E}_A}\right)\%, \qquad \bar{E}_A > 0. \tag{18}$$

The retained horizon is $N = \lceil (220/v + 2)/0.01 \rceil$. Episode failure uses $|e_y| > 2.5$ m or $|e_\psi| > 0.6$ rad; nonfinite state or command is a run-integrity failure requiring investigation, not a fabricated finite score. Completion, realized tracking, failure penalties and termination causes will be reported separately. Completion alone does not establish collision avoidance.

Secondary metrics include peak error, steering effort/rate, authority, accepted residual magnitude, rejection reasons and predicted-cost margins. Offline matched branches will compare predicted and realized cost ordering, using the same future continuation and disturbance realization. Such short-horizon diagnostics remain distinct from episode-level benefit.

Comparisons will pair methods on complete scenario/noise cells. Seed-level differences will be shown, with interval estimation appropriate to the small sample and explicit uncertainty limits.

Three policy/noise pairs do not justify treating all episodes or time samples as independent training seeds. A larger seed campaign is a possible later study with a new frozen budget; it is not implied by the bounded revision12 protocol [13,14].

4.7 Timing, stability and secondary validation

Wall-time measurements will report inference and local-loop distributions, predictor cost, jitter and missed deadlines with hardware and process configuration. Simulation age and wall inference duration will occupy separate fields. Acquisition delay and command-delivery delay must be recorded if present. Nonblocking inference removes a waiting dependency but does not remove CPU contention or scheduling risk.

Local LQR analysis will state the model and domain. Bounded residuals, switching, rate limits, lag and delays must be included before any stronger stability conclusion is attempted. If no full-system theorem is established, analytical local results and empirical robustness will remain separate. A nominal predicted improvement is not a robust safety guarantee.

Webots can provide an additional robotics-simulation environment [16]. Historical native evidence uses a legacy policy interface; the current 20-feature supervised student has not been validated there. Transfer requires feature-availability checks, units/signs, timestamps, plant/actuator calibration and a matched fresh campaign. The idealized pending-worker features identified in Section 3.4 are a specific transfer prerequisite. Physical-vehicle and embedded deployment remain subsequent validation stages.

5 拟议创新点 / Proposed Contributions

The proposed contributions are assessed against prior work and the evidence actually obtained. They are not claims of an entirely new residual-learning or safety-filtering principle.

Selective residual authority over a strong baseline. The study specifies and evaluates when a bounded learned correction improves the command actually applied relative to the same LQR baseline. A zero-authority result remains valid when useful corrections are absent.

Attribution of learning, prediction and objective effects. Matched fixed proposals, retained students and controlled gate changes separate neural contribution from selection effects. The failed PPO fine-tuning pilot is included in this account rather than hidden by comparison only with LQR.

Explicit timing and execution semantics. Timestamp preservation, held/pending actions, action-age rejection and recovery are documented at the control interface. Native feature causality and wall-time execution are treated as separate empirical questions.

Auditable research evidence. Frozen protocols, complete scenario grids, policy/normalizer hashes, failure-aware metrics and reproducible reporting support bounded conclusions. The anticipated value is a reliable integration and evaluation method; novelty and practical advantage depend on the completed comparative evidence.

6 可能问题及解决措施 / Potential Problems and Responses

Potential problem	Planned response
PPO adds no improvement	Retain the supervised checkpoint; stop the failed training path under the frozen screen.
Fixed proposals explain the hybrid gain	Compare both constants under each identical gate and report learning benefit separately.
Nominal predictor admits harmful corrections	Measure prediction error and realized cost ordering; preserve mismatch cases.
Observations use unavailable worker information	Version a causal native interface, retrain affected policies and revalidate.
Timestamp or held-packet errors	Test observation-time preservation, delivery, reuse, sequence regression and stale fallback.
Recovery oscillates or resets incorrectly	Replay gate state transitions and distinguish qualifying ticks from packet arrivals.
LQR fallback fails outside its assessed region	Retain failures and define the evaluated domain; do not assume an emergency controller.
Local computation exceeds its period	Bound optional work, measure contention and validate timeout behavior before real-time claims.
Metrics or scenario grids are incomplete	Validate all planned cells, labels and failure penalties before aggregate comparison.
Native transfer changes physical assumptions	Calibrate sensing/actuation and compare fresh trajectories before transferring conclusions.

7 应用价值 / Application Value

7.1 Path-tracking research

The framework provides an experimental way to assess learned corrections when a classical tracker is already strong. Explicit rejection reasons and authority histories help explain whether a proposal is stale, invalid or predictively unhelpful. Practical tracking value must be measured against the same baseline in a declared operating domain.

7.2 Implementation potential

The modular observation, policy, execution and supervisory interfaces can support later deployment studies. Such studies still require causal observations, calibrated sensors, suitable scheduling and actuator validation. Current simulation results do not establish road safety, production readiness or a universal inference frequency.

7.3 Reuse and economic value

Versioned configurations, documented negative results and reproducible analysis can reduce duplicated experimental work. Existing compute resources and open-source tools support a modest research budget. Hardware savings, energy savings and commercial value remain unmeasured possibilities rather than demonstrated outcomes.

8 总体安排及进度 / Research Arrangement and Schedule

The following six-month plan begins after proposal approval. Existing audits, supervised checkpoints and completed PPO pilots are prerequisites, not future work to be repeated without reason. Conditional stages follow the frozen decision tree; a failed screen redirects effort to diagnosis, reporting and transfer prerequisites rather than automatic extra training.

Period	Planned work and deliverable
Month 1	Finalize literature positioning, archive the reviewed baseline, verify equations/interfaces and catalogue existing evidence.
Month 2	Complete the frozen predictor/objective development ablation and report all gate and learning-benefit screens.
Month 3	If eligible, run the independent teacher confirmation and bounded PPO pilots; otherwise document the failed path and stop it.
Month 4	Freeze eligible configurations for reserved evaluation; address causal native observations and run separately specified interface tests.
Month 5	Assess process timing and feasible Webots transfer; complete uncertainty analysis and draft the thesis/reproduction package.
Month 6	Resolve supervisor comments, audit citations and figures, and prepare submission and defense according to university dates.

9 经费预算 / Proposed Budget

The proposed allocation is CNY 4,000, subject to institutional approval. Existing computing resources and available open-source software will be used. This allocation does not establish funding for a physical-vehicle campaign or a commercial simulator.

Expense category	CNY	Basis
Research and experiment preparation	2,500	Simulation preparation and experiment support.
Literature and document access	1,000	Materials unavailable through institutional access.
Thesis printing and binding	500	Review and submission copies.
Total	**4,000**	**Proposed allocation.**

参考文献 / References

[1] R. Rajamani. *Vehicle Dynamics and Control*. 2nd ed. Springer, 2012. DOI: 10.1007/978-1-4614-1433-9.

[2] B. D. O. Anderson and J. B. Moore. *Optimal Control: Linear Quadratic Methods*. Prentice-Hall, 1990.

[3] B. R. Kiran et al. "Deep reinforcement learning for autonomous driving: A survey." *IEEE Transactions on Intelligent Transportation Systems*, 23(6), 4909–4926, 2022. DOI: 10.1109/TITS.2021.3054625.

[4] T. Johannink et al. "Residual reinforcement learning for robot control." *IEEE International Conference on Robotics and Automation*, 6023–6029, 2019. DOI: 10.1109/ICRA.2019.8794127.

[5] J. Schulman, F. Wolski, P. Dhariwal, A. Radford and O. Klimov. "Proximal policy optimization algorithms." arXiv:1707.06347, 2017. https://arxiv.org/abs/1707.06347.

[6] J. Schulman, P. Moritz, S. Levine, M. I. Jordan and P. Abbeel. "High-dimensional continuous control using generalized advantage estimation." *International Conference on Learning Representations*, 2016. https://arxiv.org/abs/1506.02438.

[7] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus and N. Dormann. "Stable-Baselines3: Reliable reinforcement learning implementations." *Journal of Machine Learning Research*, 22(268), 1–8, 2021. https://jmlr.org/papers/v22/20-1364.html.

[8] J. P. Hespanha, P. Naghshtabrizi and Y. Xu. "A survey of recent results in networked control systems." *Proceedings of the IEEE*, 95(1), 138–162, 2007. DOI: 10.1109/JPROC.2006.887288.

[9] B. Wittenmark, J. Nilsson and M. Törngren. "Timing problems in real-time control systems." *American Control Conference*, vol. 3, 2000–2004, 1995. DOI: 10.1109/ACC.1995.531240.

[10] K. P. Wabersich and M. N. Zeilinger. "A predictive safety filter for learning-based control of constrained nonlinear dynamical systems." *Automatica*, 129, 109597, 2021. DOI: 10.1016/j.automatica.2021.109597.

[11] L. Brunke et al. "Safe learning in robotics: From learning-based control to safe reinforcement learning." *Annual Review of Control, Robotics, and Autonomous Systems*, 5, 411–444, 2022. DOI: 10.1146/annurev-control-042920-020211.

[12] Á. Domina and V. Tihanyi. "Model predictive controller approach for automated vehicle's path tracking." *Sensors*, 23(15), 6862, 2023. DOI: 10.3390/s23156862.

[13] P. Henderson et al. "Deep reinforcement learning that matters." *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), 3207–3214, 2018. DOI: 10.1609/aaai.v32i1.11694.

[14] R. Agarwal, M. Schwarzer, P. S. Castro, A. C. Courville and M. G. Bellemare. "Deep reinforcement learning at the edge of the statistical precipice." *Advances in Neural Information Processing Systems*, 34, 2021.

[15] R. Wilhelm et al. "The worst-case execution-time problem—overview of methods and survey of tools." *ACM Transactions on Embedded Computing Systems*, 7(3), Article 36, 2008. DOI: 10.1145/1347375.1347389.

[16] O. Michel. "Cyberbotics Ltd. Webots: Professional mobile robot simulation." *International Journal of Advanced Robotic Systems*, 1(1), 39–42, 2004. DOI: 10.5772/5618.

[17] LANS project. "Step3 complete: retain the supervised student; PPO adds no gain." Research record, 2026. Commit `17dbd78`. Archived report.

[18] LANS project. "Full project audit and corrections." Research record, 25 October 2026. Commit `17dbd78`. Audit record.

[19] LANS project. "Isolated predictor and objective development ablation." Frozen protocol and implementation, 2026. Commit `17dbd78`. Protocol and implementation.